

HW 2

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```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Problem 1

1a, create a simulated dataset

simulate():
generate 100 UIDs starting from 100000000
Create homework 1-5 and quiz 1-7, using floor and runif to get a randomized value between 1-100

```
gradebook <- simulate_data()
# arrange by increasing UID
gradebook <- gradebook %>%
  arrange(UID)
head(gradebook)
```

```
## # A tibble: 6 x 13
##       UID Homework_1 Homework_2 Homework_3 Homework_4 Homework_5 Quiz_1 Quiz_2
##   <int>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>  <dbl>  <dbl>
## 1 105467322      80      79      100      91      53      62      78
## 2 107401527      68      62      90      92      55      68      87
## 3 117233698      80      79      79      83      81      83      66
## 4 121937536      82      52      83      78      54      79      68
## 5 124861589      72      65      73      75      84      94      79
## 6 126100703      84      73      50      94      63      71      85
## # i 5 more variables: Quiz_3 <dbl>, Quiz_4 <dbl>, Quiz_5 <dbl>, Quiz_6 <dbl>,
## #   Quiz_7 <dbl>
```

1b) write R code in the R markdown file to randomly replace 10% of the values in HW1, HW5, and Quiz3 with NA values. Use `is.na()` and `sum()` to display the results

Algorithm:

generate random indices as a logical vector(replace or not replace)
subset the data and replace TRUE index with NA

```
gradebook$Homework_1 <- replace_10_percent(gradebook$Homework_1)
gradebook$Homework_5 <- replace_10_percent(gradebook$Homework_5)
gradebook$Quiz_3 <- replace_10_percent(gradebook$Quiz_3)
```

summary of the replacements:

```
gradebook %>%
  select(Homework_1, Homework_5, Quiz_3) %>%
  summarise(sum(is.na(.)))
```

```
## # A tibble: 1 x 1
##   'sum(is.na(.))'
##           <int>
## 1             30
```

```
cat("30 data points should be replaced as NA.")
```

30 data points should be replaced as NA.

1c) Write a function `messy_impute()` that imputes using missing values in the `gradebook`

```
# Convert gradebook to tidy
gradebook_tidy <- gradebook %>%
  pivot_longer(
    cols = -UID,
    names_to = "Assesment",
    values_to = "Score"
  ) %>%
  separate(
    col = Assesment,
    into = c("Assesment", "Number"),
    sep = "_ "
  ) %>%
  mutate(Number=as.numeric(.$Number))
```